

DEPARTMENT OF CHEMICAL ENGINEERING IIT KHARAGPUR
SEMI-ANNUAL SEMESTER EXAMINATION 2011, SPRING SEMESTER
 Subject: Instability & Pattern Formation in Thin Polymer Films (INSTABILITY)

Subject No: CH 62052

No of Students: 80

FULL MARKS: 30

TIME: 3 Hours

Closed Book, Closed Notes

General Instructions:

1. All Questions are compulsory.
2. Clearly write your name, Roll No., Subject Name (INSTABILITY), Subject Number on the Answer Book.
3. Feel free to assume any missing data with proper justifications.
4. **Be Precise with your answers. Long, redundant answers can potentially fetch zero!**

1. (a) Discuss the morphological evolution sequence in the dewetting of an ultra thin film. (2)
- (b) Why does rims form around a ruptured hole? How are the over all morphology of the dewetted structures get affected in case rim instability occurs. (1+1)
- (c) What is the role of surface tension in dewetting of an ultra thin film (1)
- (d) In contrast to dewetting of an ultra thin polymer film, you are observing the dewetting of an water film of similar thickness. What are the likely differences in the two systems? (1)
- (e) What is Kinematic boundary condition? Derive it's expression for an ultra thin film. Comment on the order of time from it (1+2+1) [10]

2. (a) The mathematical form for the lubrication approximated thin film equation is:

$$h_t - 0.33\mu^{-1} [(h^3 P_x)_x] = 0 \quad (\text{where } P_x = \partial P / \partial x \text{ etc.})$$

Find out the condition of the instability of a film based on a Linear Stability Analysis of the above equation around an initial film thickness of h_0 . Assume that the only interfacial interaction is due to non retarded van der Waals force. You can use the expression $\Delta p = \gamma h_{xx}$ but must mention the condition under which this approximation is valid. (4)

- (b) Any comment how the condition/ expression may change if you consider that the interfacial interaction is due to retarded van der Waals force. (1)
 - (c) A quartz glass plate is coming in contact with a thin polymer thin film of thickness h coated on another quartz substrate. Both the film and the substrate are submerged in a liquid medium. Find the excess free energy of the system as a function of film – contactor separation distance. Assume that the only interfacial interaction is due to non retarded van der Waals force. (4)
 - (d) Write down the expression for equilibrium contact angle for a drop resting on a solid substrate in terms of spreading coefficient. (1) [10]
3. (a) How does a film form in spin coating? (2)
 - (b) What will happen if in spin coating you gradually reduce the polymer concentration of the dispensed drop? (2)

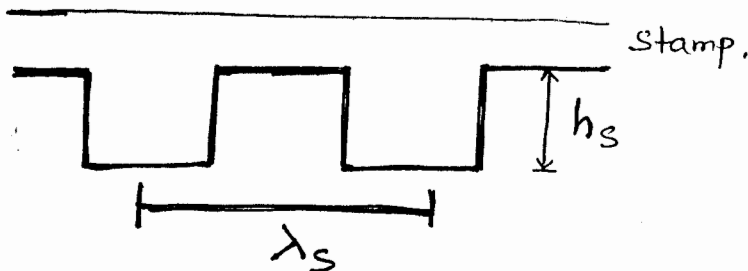
- (c) In case you are spin coating on a stripe patterned surface, what will be the morphology of the coated film? (2)
- (d) How can, in real conditions you can get films of uniform thickness on a topographically structured substrate? (1)
- (e) How does film thickness vary with dip speed in dip coating and why? (1)
4. (a) Suppose you are imprinting Sylgard 184 (crosslinkable PDMS) film with patterned stamps by imprinting. Will the imprinted patterns be permanent (justify your answer)? If not, what additional thing needs to be done to make the patterns permanent? (1+1)
- (b) Can you think of an experimental protocol for the above system by which you can create structures with the same periodicity of the stamp pattern but having feature height lower than the original height on the stamp? [Please refer to fig 1 below] (3)
- (c) Is there any limit on the magnitude of surface tension that can be measured using a contact angle goniometer (justify your answer)? (2)
- (d) What is Hydration pressure? (1)
5. (a) What are the different printing modes in photolithography? Mention one advantage of each mode. (1+1)
- (b) To which of the above mode is immersion lithography most closely associated to? (1)
- (c) What is nanosphere lithography? (1)
- (d) What is Disjoining Pressure? (2)
- (e) Please justify/ mention how a stable dispersion is possible in a liquid in terms of surface/ interfacial tension (2)
- (f) What is stick slip type motion? (1)
- (g) How do you justify the "No slip" boundary condition during the movement of a 3 phase contact line (1)
6. (a) Discuss the sequence of approach of a fully aligned AFM cantilever tip. (2)
- (b) Discuss how an AFM image is generated? (2)

[8]

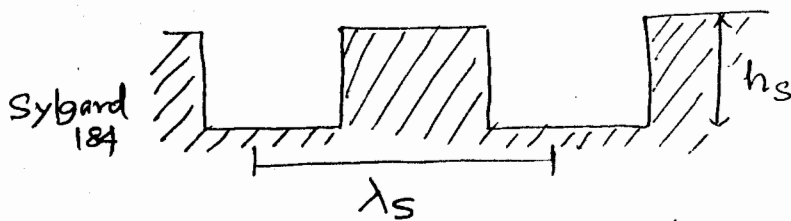
[8]

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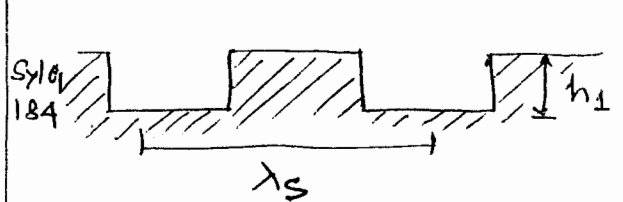
You want to make this



Perfect Negative Replica

[This is normal in any imprinting]

[Figure 1]



$h_1 < h_s$ and may want to ~~low~~ control h_1 between 0 and h_s
 $0 < h_1 < h_s$